

7.3 Purification faults

7.3.1 Unsatisfactory separation result

Possible cause	Action
Gravity disc hole too small.	Use a gravity disc with a larger hole.
Incorrect separating temperature.	Adjust temperature.
Throughput too high.	Reduce throughput.
Sludge space in bowl is filled.	Empty the sludge basket in the bowl.
Disc stack clogged.	Clean the bowl discs.
Bowl speed too low.	See “7.2.10 Speed too low” on page 119.

7.3.2 Outgoing water contaminated by oil

Possible cause	Action
Gravity disc hole too large.	Use a gravity disc with a smaller hole.
Seal ring under the gravity disc defective.	Fit a new seal ring.

7.3.3 Broken water seal

Possible cause	Action
Gravity disc too large.	Use a gravity disc with a smaller hole.
Separation temperature too low.	Increase temperature.
Throughput too high.	Reduce throughput.
Sealing water volume too small.	Supply more water.
Seal ring under gravity disc defective.	Fit a new seal ring.
Disc stack clogged.	Clean the bowl discs.
Bowl speed too low.	Use correct speed. See "7.2.10 Speed too low" on page 119.
Bowl incorrectly assembled.	Examine and make correct.

7.4 Clarification faults

7.4.1 Unsatisfactory separation result

Possible cause	Action
Separating temperature too low.	Adjust.
Throughput too high.	Reduce throughput.
Feed oil contains water.	Re-assemble and operate the separator as a purifier.
Disc stack clogged.	Clean the bowl discs.
Sludge space in bowl filled.	Empty the sludge basket.
Bowl speed too low.	See “7.2.10 Speed too low” on page 119.

7.4.2 Oil discharge through water outlet

Possible cause	Action
Valve(s) in outlet line closed.	Open the valve(s) and adjust to normal back pressure.
Disc stack clogged.	Clean the bowl discs.
Seal ring under gravity disc is defective.	Fit a new seal ring.
Bowl incorrectly assembled.	Examine and make correct.

8 *Technical Reference*

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8.1 Technical data

Alfa Laval ref. 558168

Units according to ISO Standard.

The manufacturer reserves the right to change specifications without notice.

Product number	881241-08-16	
Separator type	MAB 104B-14/24C	
Purpose	Continuous purification of mineral oil from solid particles and water. The flash point of the oil to be separated must be minimum 60 °C.	
Hydraulic capacity	Maximum 2,0 m ³ /h	
Recommended throughput	Marine Diesel Oil: viscosity 13 cSt/40 °C	0,9 m ³ /h
Max. density of sediment	1600 kg/m ³	
Max. density of feed	991 kg/m ³ , (mechanical safety max. 1100 kg/m ³)	
Feed temperature	Minimum 0 °C Maximum +100 °C	
Ambient temperature	Minimum +5 °C Maximum +55 °C	
Motor (with pump)	4-pole 1,5 kW standard motor, 50 or 60 Hz, 3-phase. Direct on-line start. If Y/D-start is used maximum 5 seconds in Y position.	
Motor (without pump)	4-pole 1,1 kW standard motor, 50 or 60 Hz, 3-phase. Direct on-line start. If Y/D-start is used maximum 5 seconds in Y position.	
Power consumption (50Hz)	idling	0,5 kW
	running (at max. capacity)	1,8 kW
	(at starting-up)	1,3 kW

Speed

The prescribed speed of the bowl spindle which must not be exceeded, is stamped on the name plate of the machine.

Gear ratio	50: 10	49: 12
Maximum speed of rotation, rpm	50 Hz	60 Hz
Motor shaft speed	1500	1800
Revolution counter speed	79 - 83	95 - 100
Bowl speed	7500	7350

Starting time

2 - 3 minutes

Stopping time

without brake

9 minutes (average)

Maximum running time without flow

empty bowl

480 minutes

filled bowl

480 minutes

Inlet pump, built-on

Capacity

50 Hz

1,08 m³/h

60 Hz

1,30 m³/h

Suction lift

Max. 4 m wg (manometric)

Delivery head

Max. 15 m wg (manometric)

Outlet pump, built-on

Capacity

50 Hz

1,19 m³/h

60 Hz

1,43 m³/h

Delivery head

15 - 25 m wg

(depending on flow and viscosity)

Sludge and water space volume

1,26 litres

Required water quality

Fresh water

Lubrication

see "8.8.1 Lubrication chart, general" on page 137

Lubricating oil volume

0,8 litres

Weight

Separator without motor

Net weight approx. 149 kg

Motor

Approx. 58kg

Complete bowl

Approx. 25 kg

Shipping data	According to "Basic equipment"	
	Weight	280 kg gross
	Volume	0,79 m ³
Sound power level	9,3 bel(A) ISO 3744	
Sound pressure level	78 dB(A) ISO 3744	
Vibration level	Max. for separator in use	9 mm/s (RMS)
Materials	Bowl spindle, lock ring	steel
	Frame, lower and upper parts	cast iron ("Centriblue" finish ¹)
	Collecting cover	silumin (grey finish ¹)
	Bowl body and hood, disc stack, gravity disc	stainless steel
	Other bowl parts	brass
	Other inlet and outlet parts	stainless steel, brass, cast iron
¹ An epoxy enamel		

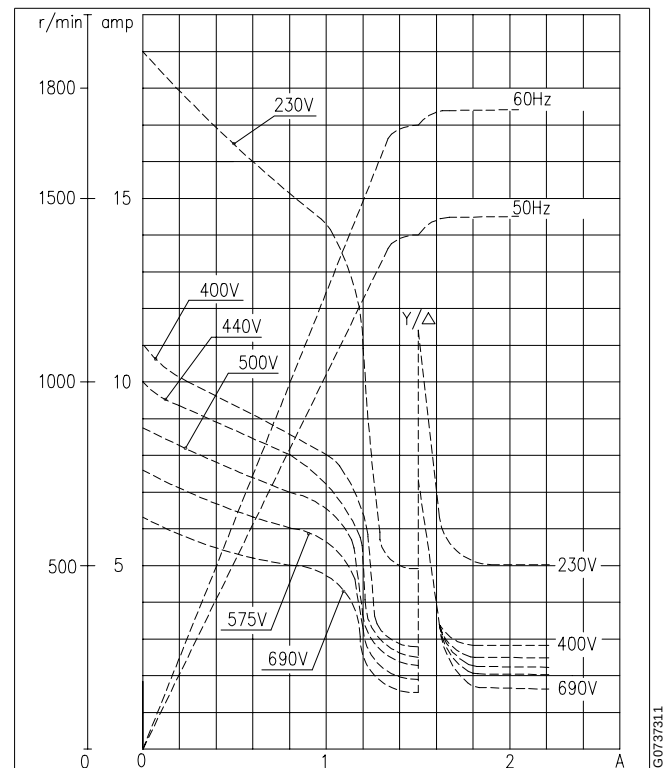
8.2 Motor drive data

Cables and fuses.

Voltage (V)	Rated current (A)	Fuse (A)	Cable size in mm ²	
			Cu	Al
230	8,3	16	2,5	4
400	4,8	10	1,5	2,5
440	4,4	10	1,5	2,5
500	3,8	10	1,5	2,5
575	3,4	10	1,5	2,5
690	2,8	10	1,5	2,5

NOTE

Recommended cable area is valid for a max. ambient temperature of 25°C and with the cables freely installed.

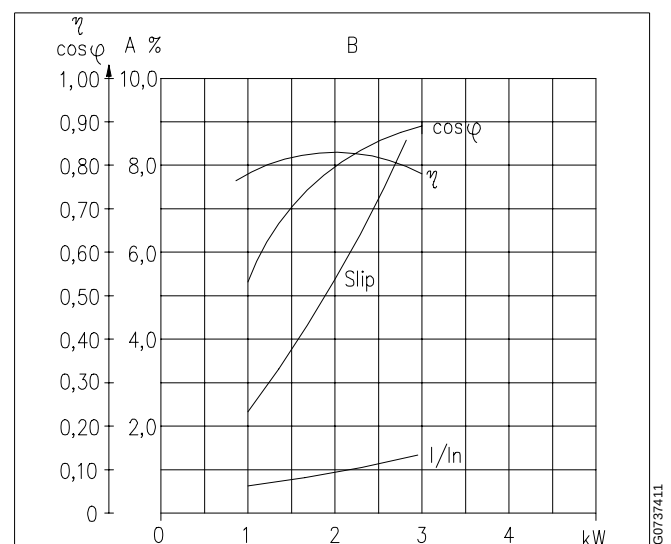


Current- and speed curves at Y/D-starting

A. Time in minutes

η and $\cos \phi$ for 60 Hz approx. of same values as for 50 Hz. Slip $\approx 0,7 \times$ slip at 50 Hz.

Bowl speed max.	7500 r/min
Motor speed min. 50 Hz	1420 r/min
Motor speed min. 60 Hz	1720 r/min
Idling power input	0,9 KW
Moment of inertia	0,187 kgm ²
Output	2,2 KW
No. of poles	4
Efficiency η	82%
Power factor $\cos \phi$	0,81
Starting torque MsY	12 Nm

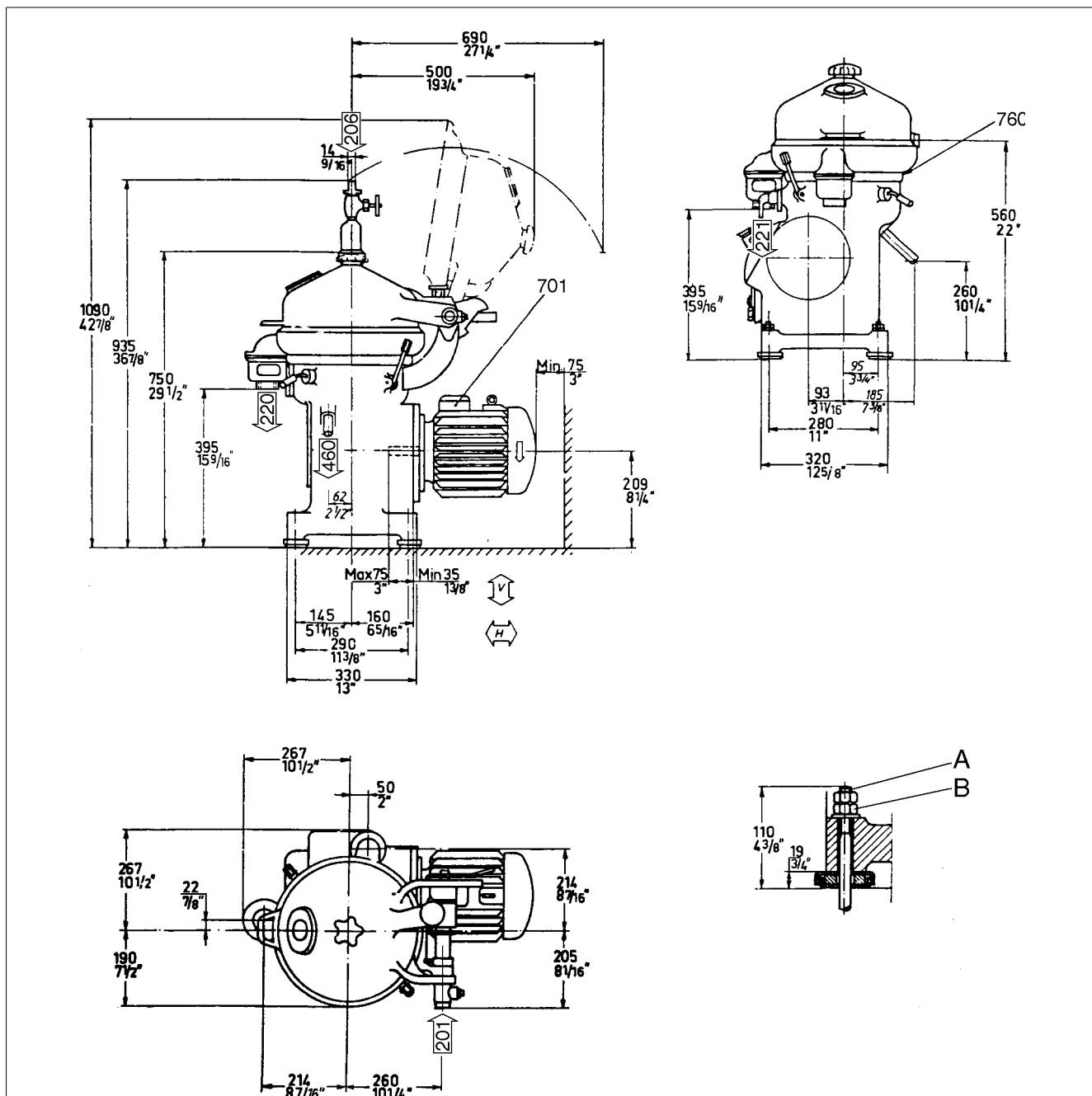


A. Slip in %

B. Performance curves

8.3 Basic size drawing

Alfa Laval ref. 558548



- A. Screw 3/8-16UNC
 B. Tightening torque 16 Nm locked with lock nut

Data for connections see "8.4 Connection list" on page 131.

All connections to be installed non-loaded and flexible.



Vertical force not exceeding
 8 kN/foot



Horizontal force not
 exceeding 8 kN/foot

8.4 Connection list

Alfa Laval ref. 558528

Connection No.	Description	Requirements/limit
201	Inlet for process liquid - Allowed temperatures	min. 0 °C, max. 100 °C
206	Inlet for water seal	Fresh water, approx 2,5 litres (depending on gravity disc)
220	Outlet for light phase (oil from pump)	no counterpressure
221	Outlet for heavy phase (water)	no counterpressure
460	Drain of frame	
701	Motor for separator - Max. deviation from nominal frequency (momentarily during maximum 5 seconds)	±5% ±10%
760	Cover interlocking switch (delivered as option) Type: Double, two-way microswitch Switch rating: Voltage Current	Max. 500 V Max. 3 A

Alfa Laval ref. 558549



- Data for connections see “8.6 Connection list, for heater” on page 133.

8.6 Connection list, for heater

Alfa Laval ref. 558529

Connection No.	Description	Requirements/limit
201.1	Inlet for process liquid, (to pump) - Allowed temperatures	min. 0 °C, max. 100 °C
201.2	Outlet for process liquid, (from pump to heater)	
201.3	Inlet for process liquid, (from heater to separator)	
206	Inlet for water seal	Fresh water, approx 2,5 litres (depending on gravity disc)
220	Outlet for light phase (oil from pump)	no counterpressure
221	Outlet for heavy phase (water)	no counterpressure
460	Drain of frame	
701	Motor for separator - Max. deviation from nominal frequency (momentarily during maximum 5 seconds)	±5% ±10%
760	Cover interlocking switch (delivered as option) Type: Double, two-way microswitch Switch rating: Voltage Current	Max. 500 V Max. 3 A

8.7 Interface description

Alfa Laval ref. 558999

In addition to the Connection List this document describes limitations and conditions for safe control, monitoring and reliable operation.

At the end of the document a function graph and running limitations are to be found.

8.7.1 Definitions

Standstill (Ready for start) means:

- The machine is assembled correctly.
- All connections are installed according to Connection List, Interconnection Diagram and Interface Description.

Start means:

- The power to the separator is on.
- The acceleration is supervised to ensure that a certain speed has been reached within a certain time. See "8.1 Technical data" on page 126.

The start procedure continues until the full speed has been reached and a stabilization period has passed (about 1 minute).

Normal stop means:

- Stopping of the machine at any time with brake applied. The bowl must be kept filled.

Safety stop means:

The machine must be stopped in the quickest and safest way due to vibrations or process reasons.

Comply to following conditions:

- The bowl must be kept filled.
- The machine must not be restarted before the reason for the Safety stop has been investigated and action has been taken.

In case of emergency condition in the plant, the machine must be stopped in a way that is described in EN 418.

8.7.2 Component description and Signal processing Electrical Connections

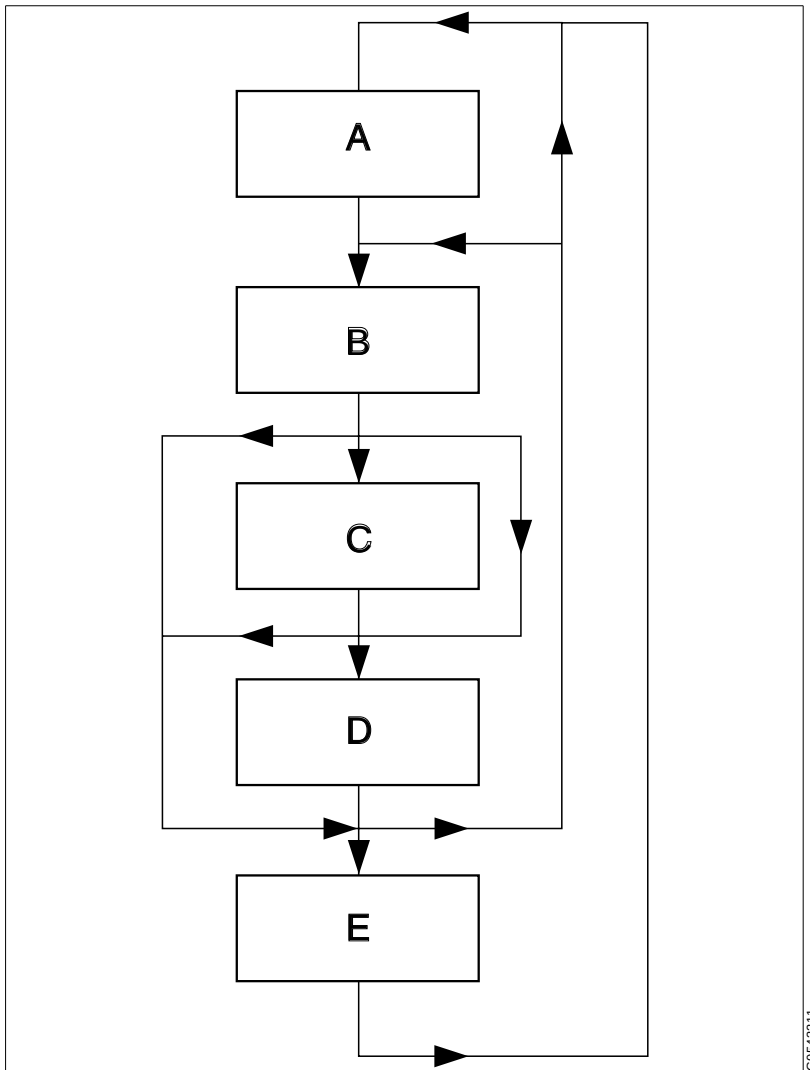
Separator motor 701

The separator is equipped with a 3-phase Y-D started. The motor is of control torque type and built for a long starting time. The starting equipment must be dimensioned for twice the rated current of the motor, and the overload relay must only be connected in the D-line.

Cover interlocking switch 760 (Option)

The cover of the separator is equipped with an interlocking switch. When the cover is closed the interlocking circuit in the control system is closed and the separator could be started.

8.7.3 Function graph and running limitations



- A. Stand still
- B. Starting mode
- C. Running mode
- D. Stop mode
- E. Safety stop mode

8.8 Lubricants

8.8.1 Lubrication chart, general

Alfa Laval ref. 553216-01

Lubricating points	Lubricants
Bowl spindle ball bearings and buffers are lubricated by oil mist.	Lubricating oil as specified in "8.8.2 Recommended lubricating oils" on page 139.
Bowl spindle taper.	Lubricating oil (only a few drops for rust protection).
Buffers of bowl spindle.	Lubricating oil.
Bowl: Sliding contact surfaces and pressure loaded surfaces such as lock rings, threads of lock rings, bowl hood, and cap nut.	Pastes as specified in "8.8.4 Recommended lubricants" on page 143. If not specified otherwise, follow the supplier's recommendation about method of application.
Rubber seal rings.	Grease as specified in "8.8.4 Recommended lubricants" on page 143.
Friction coupling ball bearings.	The bearings are packed with grease and sealed and need no extra lubrication.
Electric motor.	Follow manufacturer's instructions.

Alfa Laval Lubricating Oil Groups:

- **Group A oil:** a high quality gear oil on paraffin base with stable AW (anti wear) additives.
- **Group B oil:** a high quality gear oil on paraffin base with stable EP (extreme pressure) additives.
- **Group D oil:** a synthetic base oil with additives stable at high operating temperatures.

Do not mix different oil brands or oils from different oil groups.

Always use clean vessels when handling lubricating oil.

Great attention must be paid not to contaminate the lubricating oil. Of particular importance is to avoid mixing of different types of oil. Even a few drops of motor oil mixed into a synthetic oil may result in severe foaming.

Any presence of black deposits in a mineral type oil is an indication that the oil base has deteriorated seriously or that some of the oil additives have precipitated. Always investigate why black deposits occurs.

If it is necessary to change from one group of oil brand to another it is recommended to do this in connection with an overhaul of the separator. Clean the gear housing and the spindle parts thoroughly and remove all deposits before filling the new oil.

NOTE

Always clean and dry parts (also tools) before lubricants are applied.

NOTE

Check the oil level before start. Top up when necessary. Oil volume see "8.1 Technical data" on page 126.

It is of utmost importance to use the lubricants recommended in our documentation. This does not exclude, however, the use of other brands, provided they have equivalently high quality properties as the brands recommended. The use of oil brands and other lubricants than recommended, is done on the exclusive responsibility of the user or oil supplier.

Applying, handling and storing of lubricants

Always be sure to follow lubricants manufacturer's instructions.

8.8.2 Recommended lubricating oils

Alfa Laval ref. 553219-10

Three different groups of lubricating oils are approved. They are designated as Alfa Laval lubricating oil groups A, B and D.

The numerical value after the letter states the viscosity grade.

The corresponding commercial oil brands according to "Alfa Laval lubricating oil group A/220" on page 140.

Ambient temperature °C	Alfa Laval lubricating oil group	Time in operation Oil change interval
between +5 and +45	A/220B/220	1500 h
between ±2 and +65	D/220	2000 h

Note:

- In a new installation or after change of gear transmission, change oil after 200 operating hours.
- When the separator is operated for short periods, lubricating oil must be changed every 12 months even if the total number of operating hours is less than stated in the recommendations above.
- Check and prelubricate spindle bearings on separators which have been out of service for 6 months or longer.
- In seasonal operation: change oil before every operating period.

8.8.3 Recommended oil brands

Alfa Laval lubricating oil group A/220

Alfa Laval ref. 553218-01

Viscosity grade VG (ISO 3448/3104) 220

Viscosity index VI (ISO 2909) > 92

Manufacturer	Designation
Castrol	Alpha ZN 220
Esso Standard Oil/ Svenska Statoil/Exxon	Teresso 220 Terrestic 220
Lubmarine/Beijer (ELF Brand designation according to ELF)	Polytelis 220
Mobil	DTE Oil BB
Nynäs	TD EX 220
Optimol Ölwerke	Ultra 220
Q8/Kuwait (Gulf)	Harmony AW 220
Shell	Tellus C 220 Tellus 220
Texaco/Caltex	Regal R&O 220 Paper Machine HD 220

Alfa Laval lubricating oil group B

Alfa Laval ref. 553218-02

Viscosity grade VG (ISO 3448/3104) 220

Viscosity index VI (ISO 2909) > 92

Manufacturer	Designation
Bel-Ray	100/06-220
BP	Energol GR-XP 220
Castrol	Alpha SP 220
Chevron	Ultragear 220 (NL Gear Compound 220)
DIN 51517, part 3	CLP 220
Esso Standard Oil/ Svenska Statoil/Exxon	Spartan EP 220
Fina/Petrofina	Giran 220
ISO/DIS 6743/6	L-CKC-220
Lubmarine/Beijer (ELF Brand designation according to ELF)	Epona Z 220
Mobil	Mobilgear 630 (Mobilgear SHC 220)
Nynäs	(GL 220)
Optimol Ölwerke	Optigear BM 220 (5150)
Q8/Kuwait (Gulf)	Goya 220
Shell	(Lorina 220) Omala 220 Delima HT 220
Texaco	Meropa 220
Soviet (GHOST-) Standard	IGP-152 according to TU 38.101413-90

Alfa Laval lubricating oil group D / 220*Alfa Laval ref. 553218-03*

Viscosity grade VG (ISO 3448/3104) 220

Viscosity index VI (ISO 2909) > 135

Manufacturer	Designation
Alfa Laval	542690-80 20 litres 542690-81 4 litres 542690-82 208 litres 542690-83 1 litres
BP	Energol HTX 220
Castrol	Alpha Synt T220
Lubmarine/Beijer (ELF Brand designation according to ELF)	Epona SA 220
Mobil	SHC 630
Shell	Delima HT 220 Paolina 220

8.8.4 Recommended lubricants

Pastes and bonded coatings for non-food applications

Alfa Laval ref. 553217-01

Manufacturer	Designation	Alfa Laval No.	Application
Gleitmolybdän	Gleitmo 805 K or 805 K varnish 901 Gleitmo Paste G rapid	537086-04	All pressure loaded surfaces
Dow Corning	Molykote paste 1000 spray D321 R varnish D321 R	537086-02 535586-01 535586-02	
Rocol	Antiscuffing paste (ASP)		
Klueber	Wolfracoat C paste		
Russian Standard	VNII NP 232 Gost 14068-90		

Silicone grease

Alfa Laval ref. 553217-01

Manufacturer	Designation	Alfa Laval No.
Dow Corning	Molykote 111 compound 100 gr 25 gr	539474-02 539474-03
Gleitmolybdän	Silicone paste 750	
Wacker	Silicone Paste P (vacuum paste)	

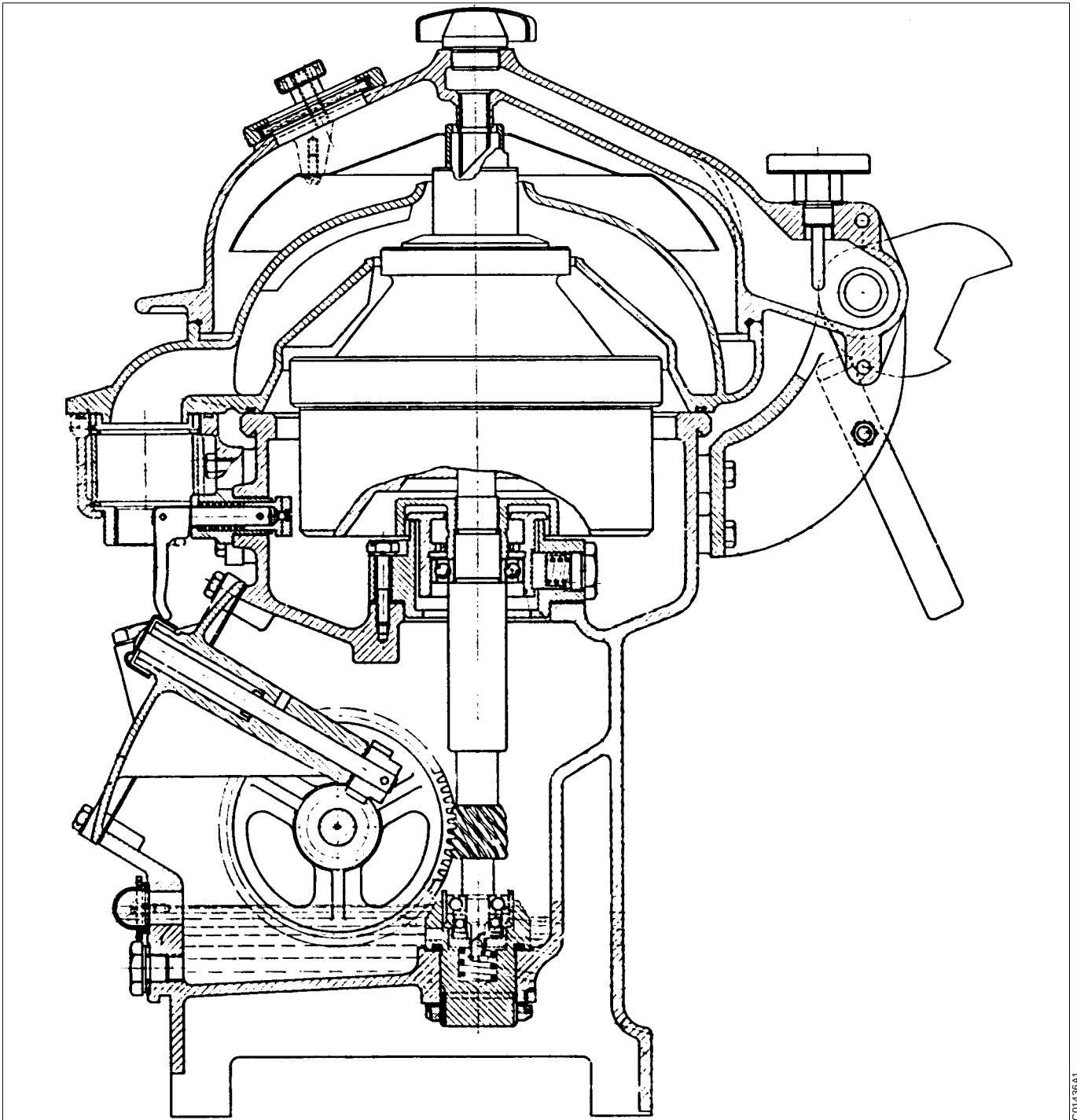
Greases for ball and roller bearings

Alfa Laval ref. 553217-01

Manufacturer	Designation	Alfa Laval No.
BP	Energrease MMEP2 Energrease LS2	
Castrol	Spheerol SW2 EP Spheerol EPL2	
Chevron	Duralith grease EP2	
Exxon	Beacon EP2	
Mobil	Mobilith SHC 460 Mobilux EP2	
Gulf	Gulflex MP2	
Q8	Rembrandt EP2	
Shell	Cailithia EP Grease T2 Alvania EP Grease 2 or R.A	
SKF	LGEP2 or LGMT2	
Texaco	Multifak AF B2 Multifak premium 2.3	
Russian Standard	Fiol 2M, Litol 24 TU 38.201.188	

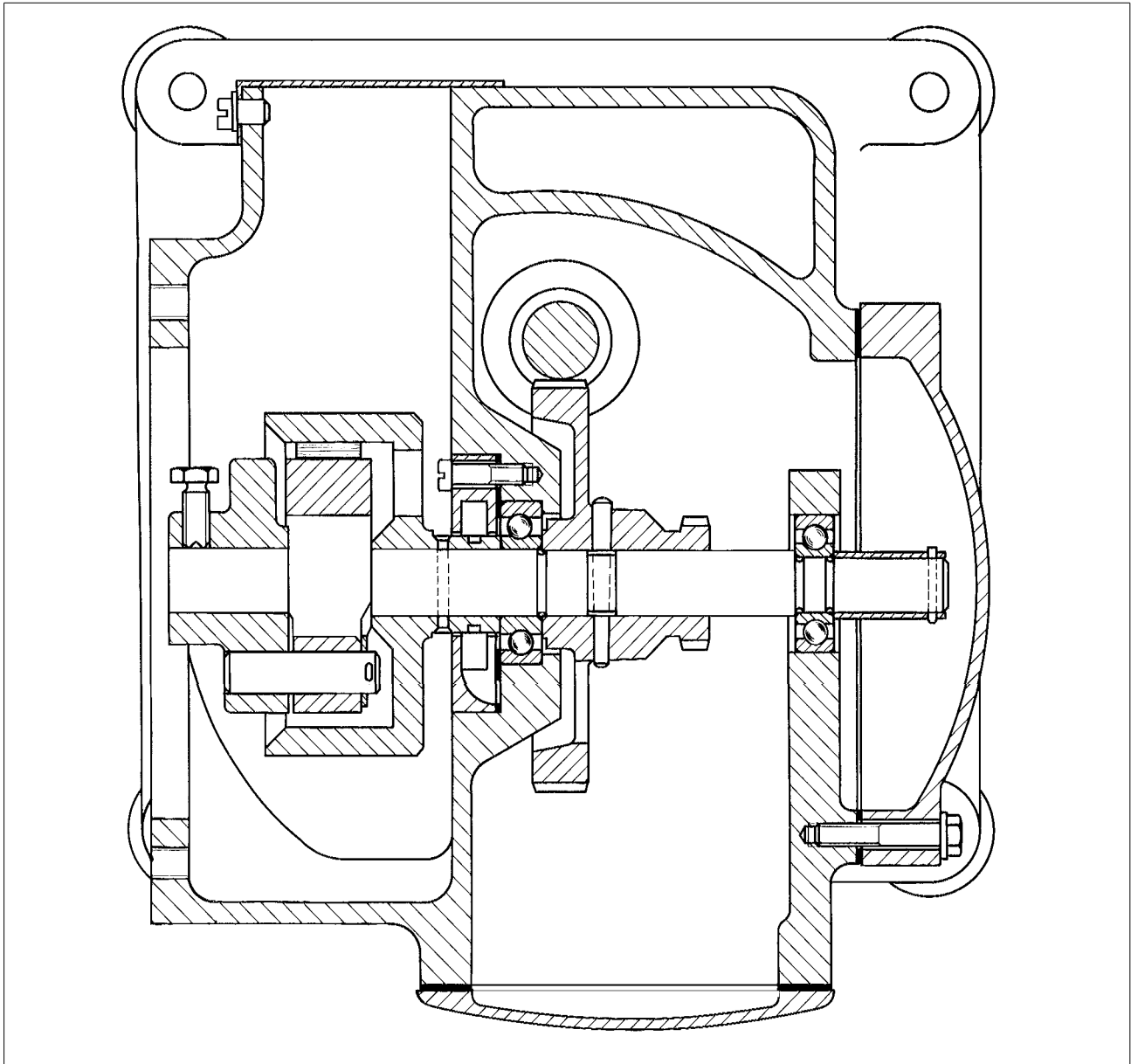
8.9 Drawings

8.9.1 Cross-section



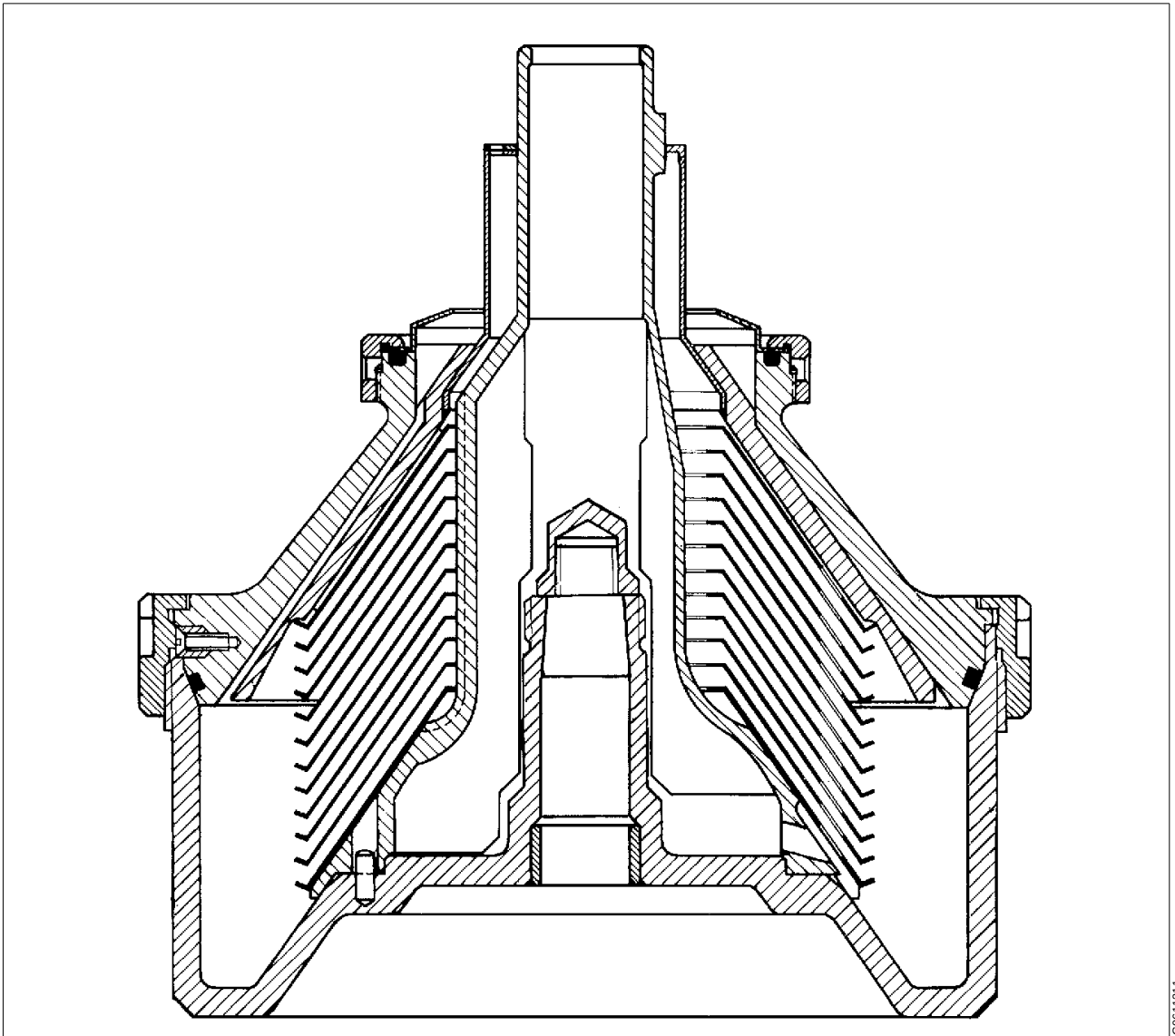
Cross-section of separator

C01436A1

8.9.2 Cross-section, frame

Cross-section, horizontal driving device

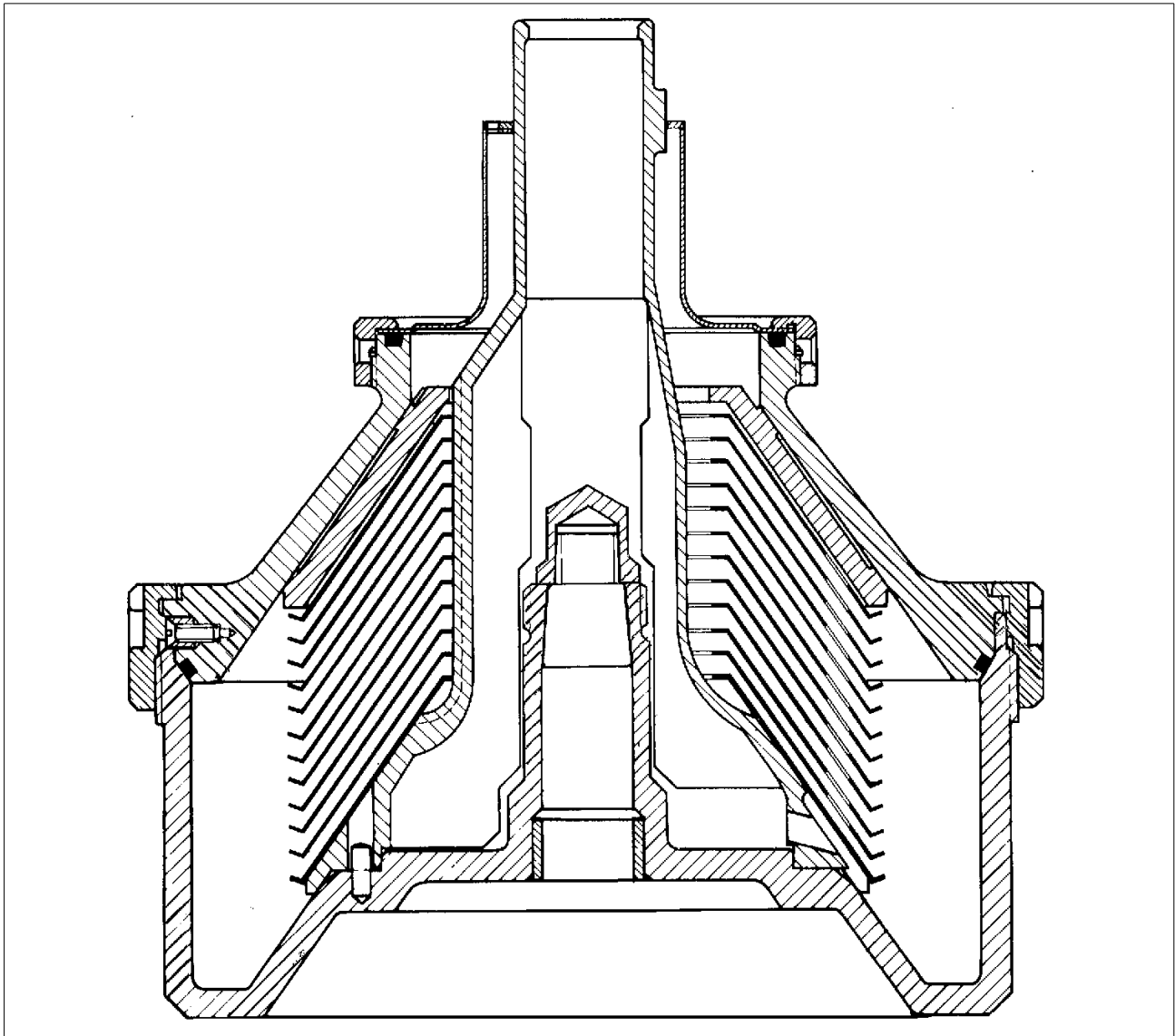
8.9.3 Purifier bowl, cross-section



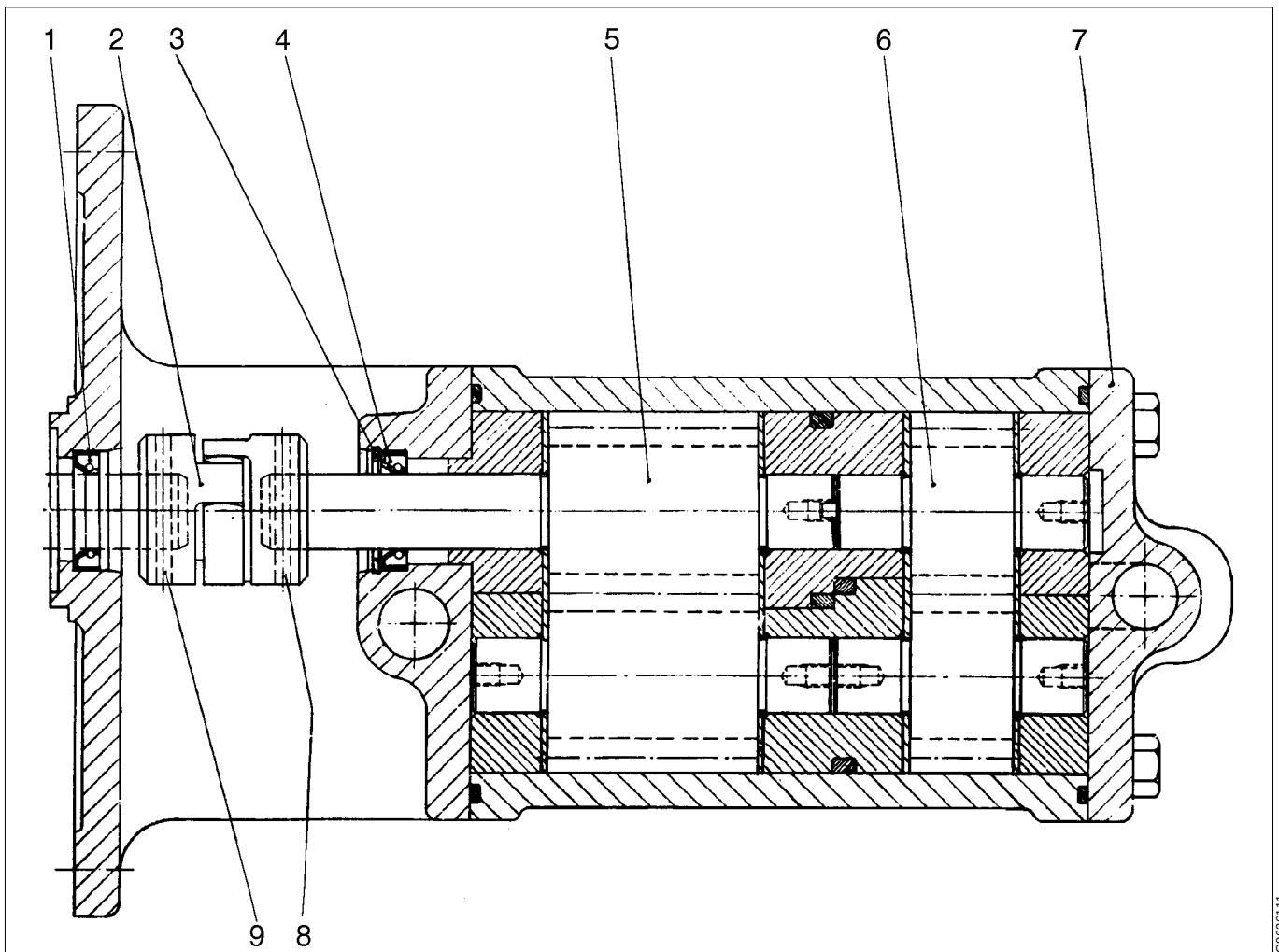
Purifier bowl

G0611211

8.9.4 Clarifier bowl, cross-section



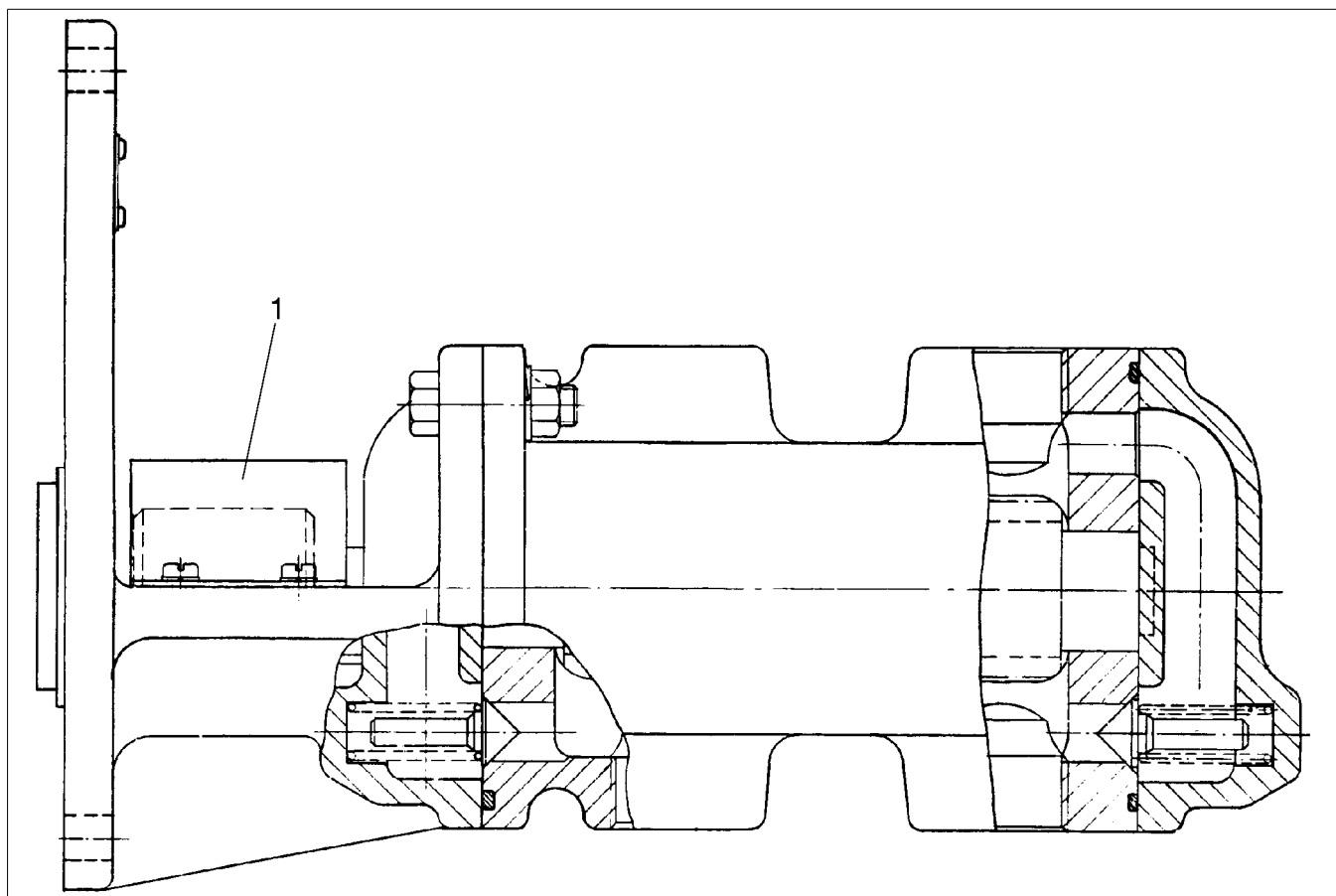
Clarifier bowl

8.9.5 Pump, cross-section

1. Sealing ring
2. Coupling
3. Retaining ring
4. Sealing ring
5. Pump impeller
6. Pump impeller
7. Shield
8. Slotted tubular spring pin
9. Slotted tubular spring pin

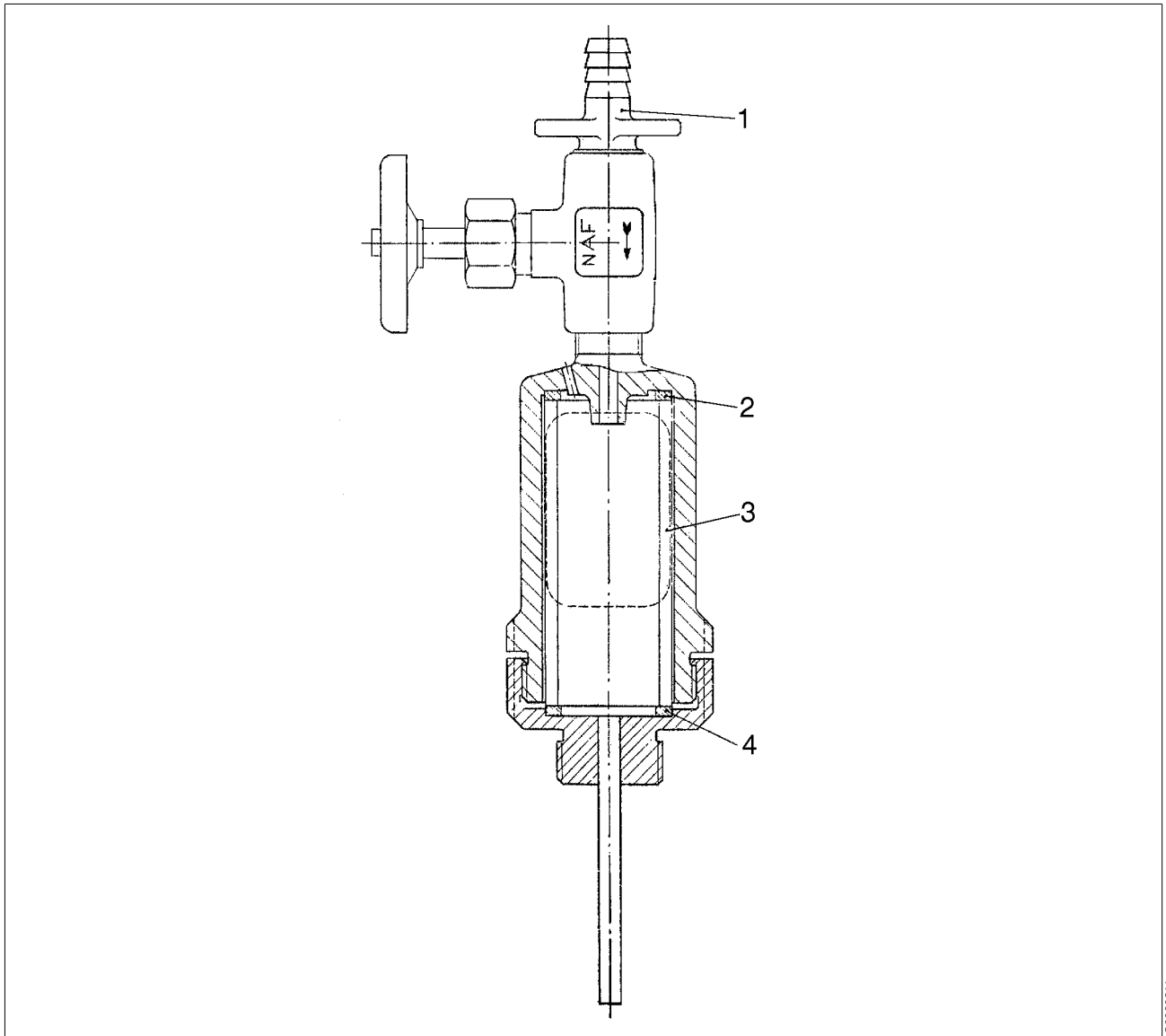
G0026111

8.9.6 Pump, cross-section



1. Coupling guard

G0626211

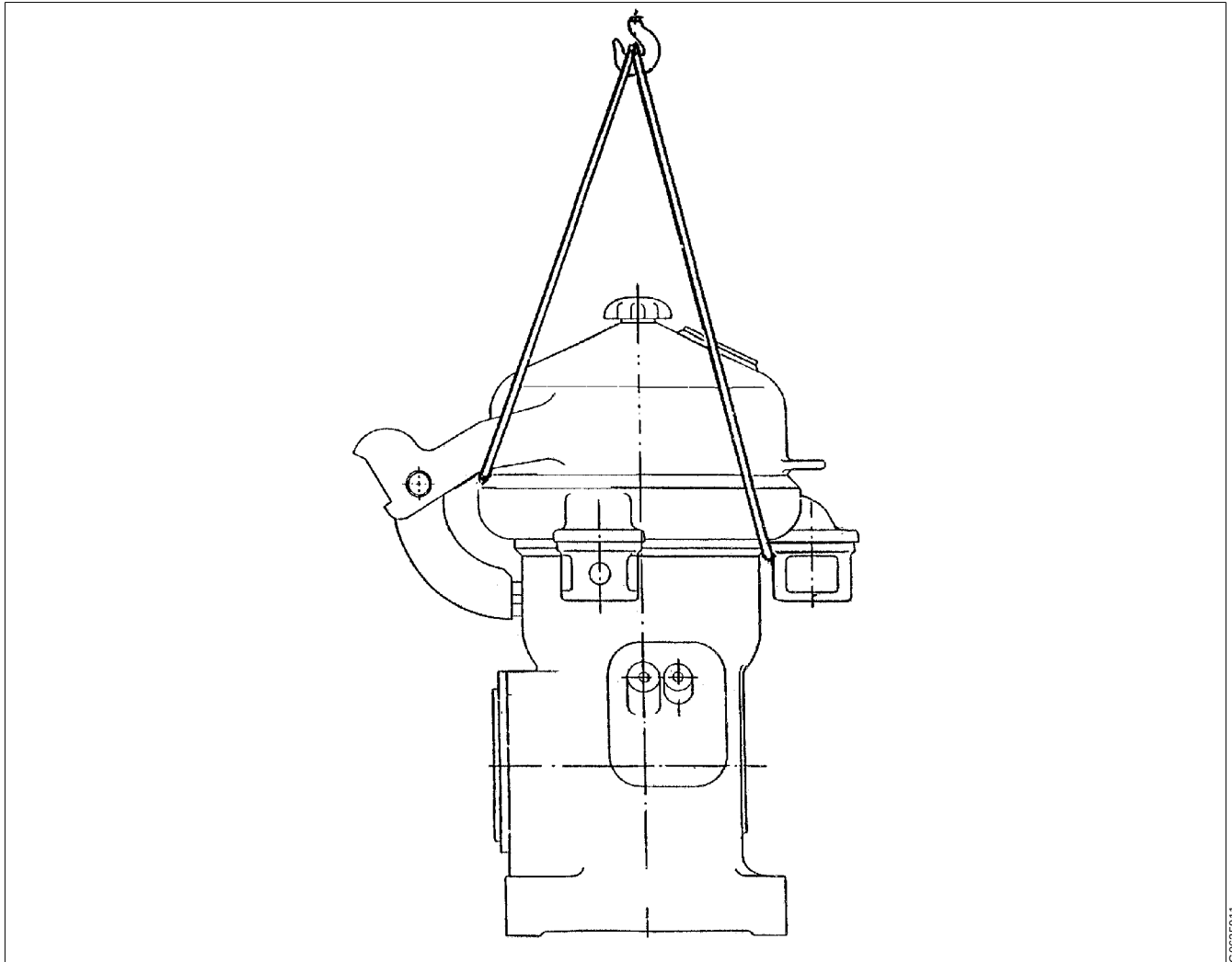
8.9.7 Water and inlet device

1. Hose nipple
2. Packing
3. Sight glass
4. Packing

G0626311

8.9.8 Lifting instruction

Alfa Laval ref. 558102



Weight to lift 200 kg

NOTE

Never lift the separator with the separator bowl inside.

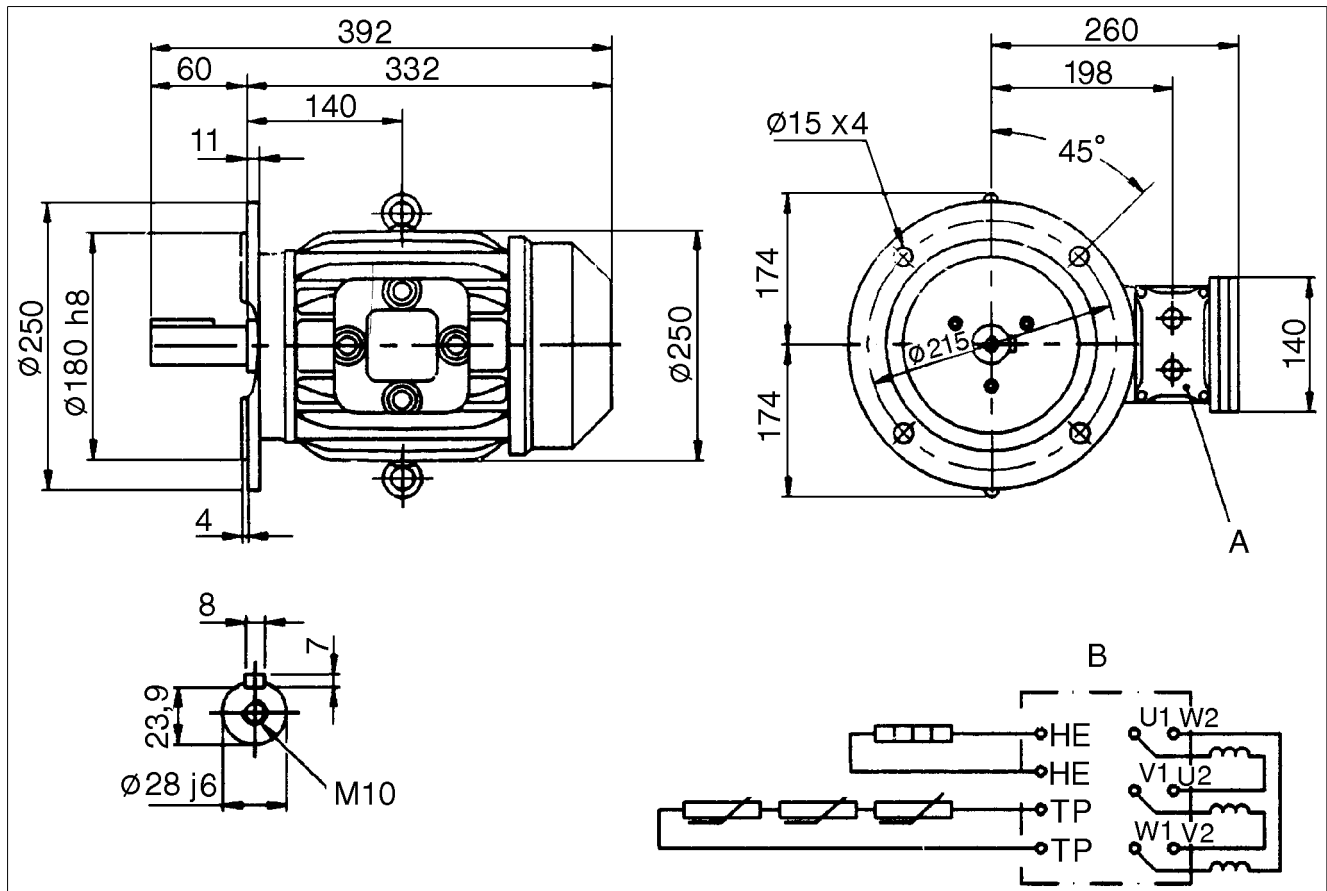
NOTE

Never lift the separator in any other way.

G0625911

8.9.9 Electric motor, Brook

Alfa Laval ref. 558487



A = Pg 16 (DIN 40430), Max wire area 2,5 mm²

B = Built in space heaters and terminal markings in main terminal box

Manufacturer	BROOK Crompton
Manufacturers drawing	H3 AGT B8554
Standards	IEC 34-1, 79-1
Size	112M
Type	AE 112 MD
Weight	58 kg
Poles	4
Insulation class	F
Bearings	DE 6306 C3 - 2Z/C3 NDE 6305 - 2Z/C3
Method of cooling	IC 0141 (IEC 34-6)
Specification	Explosion proof 3-phase CT motor for star-delta starting ¹⁾

- 1) Motor with flame proof enclosure. Certification EEx d IIB T4 according to CENELEC European Standards EN 50014 and EN 50018. Equipped with external and internal earthing.

Article No	Output kW	Speed RPM	Freq Hz	Voltage V	Current A	Pow.fac cos φ	$I_{st} / I^{1)}$	Therm. ²⁾ °C	Note
558487-01	2,2	1410	50	380 D	4,9	0,83	2,1	190	
558487-02	2,2	1420	50	400 D	4,8	0,81	2,3	190	
558487-03	2,2	1425	50	415 D	4,7	0,79	2,4	190	
558487-04	2,2	1420	50	440 D	4,4	0,81	2,3	190	
558487-05	2,2	1420	50	500 D	3,8	0,81	2,3	190	
558487-06	2,2	1410	50	660 D	2,8	0,83	2,1	190	
558487-07	2,2	1420	50	690 D	2,8	0,81	2,3	190	
558487-08	2,2	1410	50	220 D	8,5	0,83	2,1	190	
558487-09	2,2	1420	50	230 D	8,3	0,81	2,3	190	
558487-10	2,2	1720	60	380 D	5,1	0,81	2,3	190	
558487-11	2,2	1720	60	440 D	4,4	0,81	2,3	190	
558487-12	2,2	1725	60	460 D	4,3	0,78	2,4	190	
558487-13	2,2	1720	60	480 D	4,0	0,81	2,3	190	
558487-14	2,2	1720	60	575 D	3,4	0,81	2,3	190	CSA-plated
558487-15	2,2	1720	60	220 D	8,8	0,81	2,3	190	
558487-16	2,2	1725	60	230 D	8,6	0,78	2,4	190	

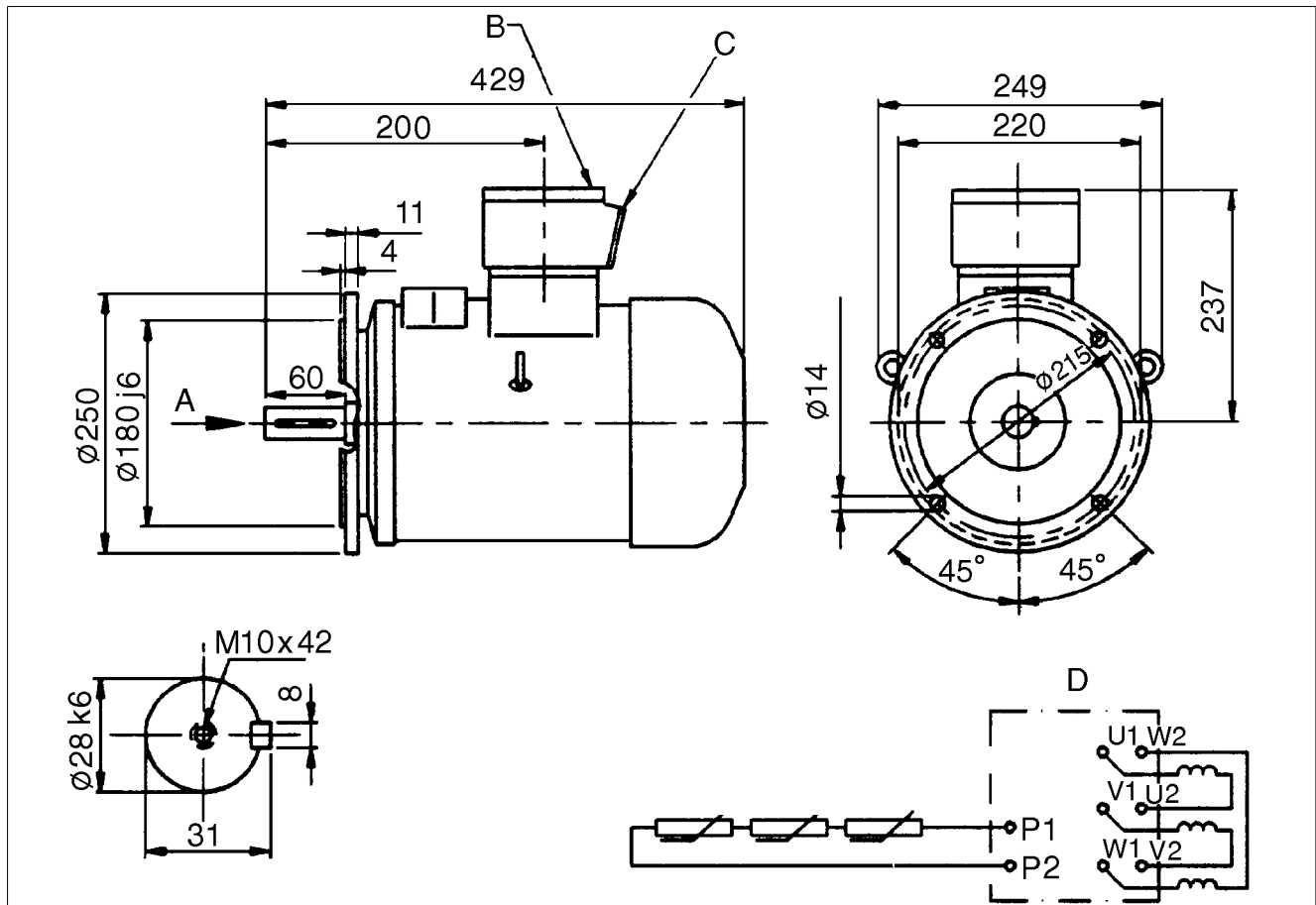
1) I_{st} / I = starting current / rated current at direct on-line starting.

2) Thermistors tripping temp.

Type of mounting: ICE 34-7		Degree of protection: ICE 34-5
	IM 1001	IP 55
	IM 3081	
	IM 3011	
	IM 3031	

8.9.10 Electric motor, LOHER

Alfa Laval ref. 558780



A = Thread M10 x 42mm

B = Terminal box rotatable at 90° intervals

C = Cable glands acc. to customers requirement. Standard design: cable entries thread 2xM25x1,5 + 1xM20x1,5

D = Terminal markings in main terminal box

Manufacturer	FLENDER ATB-LOHER
Manufacturers drawing	M 10801
Standards	IEC 34-1, 79-1, 79-7
Size	112M
Type	DWGW 112MB-04C
Weight	57 kg
Poles	4
Insulation class	F
Bearings	DE 6303-2Z/C3 NDE 6306-2Z/C3
Method of cooling	IC 0141 (IEC 34-6)
Specification	Explosion proof 3-phase CT motor for star-delta starting ¹⁾

¹⁾ Flameproof motor with increased safety terminal box. Certification EEx de IIC t4, EN 50018 and EN 50019. Equipped with external and internal earthing. PTB No Ex 82/3270. Note! Motor with special rotor HS5.

Article No	Output kW	Speed RPM	Freq Hz	Voltage V	Current A	Pow.fac cos φ	$I_{st} / I^{1)}$	Therm. ²⁾ C°	Note
558780-01	2,2	1456	50	380 D	5,2	0,79	2,5	130	
558780-02	2,2	1460	50	400 D	5,1	0,77	2,7	130	
558780-03	2,2	1463	50	415 D	5,0	0,75	2,8	130	
558780-04	2,2	1460	50	440 D	4,6	0,77	2,7	130	
558780-05	2,2	1460	50	500 D	4,1	0,77	2,7	130	
558780-06	2,2	1463	50	660 D	3,0	0,79	2,5	130	
558780-07	2,2	1460	50	690 D	2,9	0,77	2,7	130	
558780-08	2,2	1456	50	220 D	9,0	0,79	2,5	130	
558780-09	2,2	1460	50	230 D	8,9	0,77	2,7	130	
558780-10	2,2	1760	60	200 D	10,2	0,77	2,7	130	
558780-11	2,2	1760	60	440 D	4,7	0,77	2,7	130	
558780-12	2,2	1764	60	460 D	4,6	0,75	2,9	130	
558487-13	2,2	1760	60	480 D	4,3	0,77	2,7	130	
558780-14	2,2	1760	60	220 D	9,4	0,77	2,7	130	
558780-15	2,2	1764	60	230 D	9,2	0,75	2,9	130	
558780-16	2,2	1760	60	575 D	3,6	0,77	2,7	130	

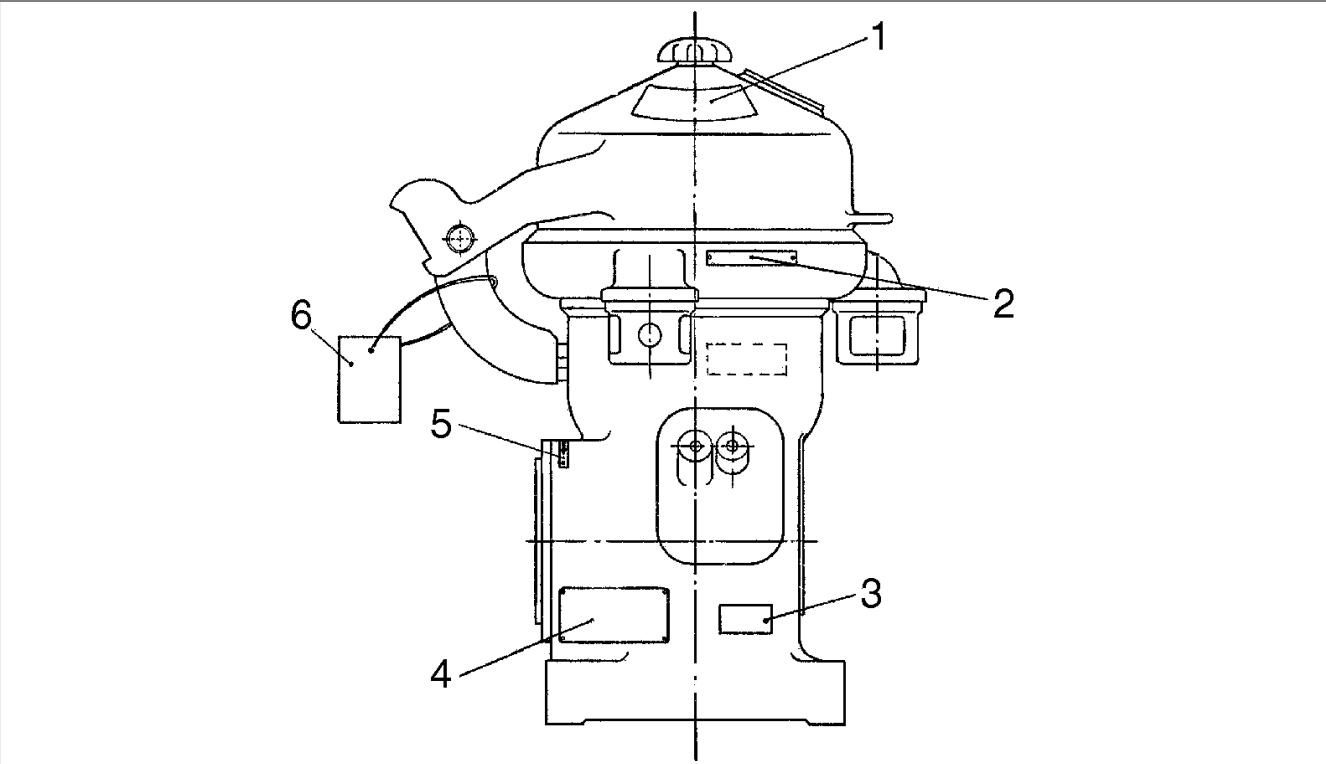
1) I_{st} / I = starting current / rated current at direct on-line starting.

2) Thermistors tripping temp.

Type of mounting: ICE 34-5		Degree of protection: ICE 34-7
	IM 1001	IP 55
	IM 3081	
	IM 3011	
	IM 3031	

8.9.11 Machine plates and safety labels

Alfa Laval ref. 558097



S0061411

5. Machine plate

Separator	MAB 104B-14/24C
Manufacturing serial No / Year	XXXX
Product No	881241-08-16
Frame	549807-05/06/07/08 (50/60 Hz with pump, 50/60 Hz)
Bowl	524724-92/82
Max. speed (bowl)	7500 r/min. (50 Hz), 7350 r/min. (60 Hz)
Direction of rotation (bowl)	←
Speed motor shaft	1500 r/min. (50 Hz), 1800 r/min. (60 Hz)
El. current frequency	50/60 Hz
Recommended motor power	2.2 kW
Max. density of feed	1100 kg/m ³
Max. density of sediment	1600 kg/m ³
Process temperature min./max.	0 / 100 °C

1. Safety label

Text on label:

DANGER

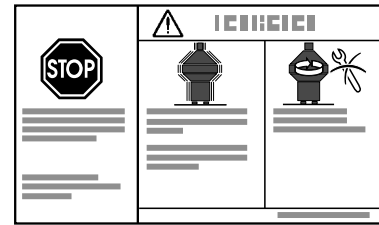
Read the instruction manuals **before** installation, operation and maintenance. Consider inspection intervals.

Failure to strictly follow instructions can lead to fatal injury.

If excessive vibration occurs, **stop** separator and **keep bowl filled** with liquid during rundown.

Out of balance vibration will become worse if bowl is not full.

Separator must **stop rotating** before **any** dismantling work is started.



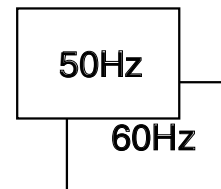
S0061521

2. Name plate



S0063211

3. Power supply frequency



S0063111

4. Arrow

Indicating direction of rotation of horizontal driving device.

6. Stop, follow lifting instruction

This transport label is not permanently fixed to the separator.

8.10 Storage and installation

8.10.1 Storage and transport of goods

Storage

Specification

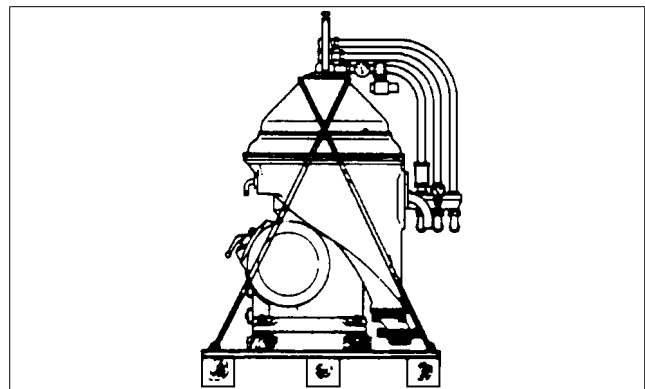
Upon arrival to the store, **check all components and keep them:**

1. Well stored and protected from mechanical damage and theft.
2. Dry and protected from rain and humidity.
3. Organized in the store in such a way that the goods will be easily accessible when installation is about to take place.

A separator can be delivered with different types of protection:

- Fixed on a pallet.

The separator must be stored in a storage room well protected from mechanical damage and theft and also dry and protected from rain and humidity.



Fixed on a pallet